

FinCompass User Guide

Rajeev Yadav, Ph.D.

August 2026

A Local Workbench for Research, Forecasting, Model Governance and Quantitative Analytics

Who this guide is for. If you want a probability for a named event, a place to watch that probability live, and a separate desk of valuation and risk tools, start at the front. If you need to audit the model, skip to Research Details and the appendices.

What this is not. FINCOMPASS is not a brokerage, not a signal service, and not a promise that a probability will be right on the next name you type. The software is built so a weak-but-valid model can still speak, provided it is labelled honestly.

How the pages are arranged. Sections 1–16 are the working product: ticker in, Forecast out, Live on, data and model maintenance. The remaining sections are the research and audit layer: scoring rules, locked tests, regimes, ensembles, troubleshooting and worked examples.

Contents

1	What FinCompass Is	6
2	The Normal Workflow	6
3	First Use: Entering a Ticker	6
4	Choosing a Forecast Period	7
5	Understanding the Benchmark	7
6	Reading the Forecast Card	7
7	Evidence Labels	8
7.1	Limited Evidence	8
7.2	Research Validated	8
7.3	Market Validated	8
8	Why a Weak Model Can Still Produce a Probability	8
9	Start Live	8
9.1	Tracking-Only Live	8
9.2	Adaptive-Eligible Live	8

10 How to Read Live	9
11 Data Updates	9
12 Three Kinds of Freshness	10
12.1 Instrument-Data Freshness	10
12.2 Model-Training Freshness	10
12.3 Retrainability	10
13 Model Maintenance	10
14 Why Labels Must Mature	10
15 The Evidence Workspace	11
16 Performance Analytics	11
16.1 Annualized Return	11
16.2 Volatility	11
16.3 Sharpe Ratio	11
16.4 Sortino Ratio	11
16.5 Maximum Drawdown	12
16.6 Beta and Tracking Error	12
17 Risk Analytics	12
17.1 Value at Risk	12
17.2 Expected Shortfall	12
17.3 EWMA Volatility	12
18 Technical Analytics	12
19 Financial Statements and Normalization	12
20 Financial Ratios	13
20.1 Profitability	13
20.2 Liquidity	13
20.3 Leverage	13
20.4 Efficiency	13
21 Discounted Cash Flow Valuation	13
22 Fixed-Income Analytics	14
22.1 Bond Price	14
22.2 Duration	14
23 Options Analytics	14

24 Portfolio Analytics	14
25 Factor and Regression Analytics	15
26 The Bayesian Reference Model	15
27 Calibration in Practical Terms	15
28 Brier Score and Log Loss	15
29 ROC AUC	16
30 Why the Test Is Locked	16
31 Regime-Aware Bayesian Research Model	16
32 Enhanced Model	16
33 Current Reference Evidence	16
34 Applicability Checks	17
35 Model Registry and Integrity	17
36 Candidate Replacement	17
37 What Happens When an Update Fails	18
38 Troubleshooting: Forecast Unavailable	18
38.1 Not Enough Ticker History	18
38.2 Benchmark History Missing	18
38.3 No Applicable Model Family	18
38.4 Model Artifact Failed Integrity Checks	18
38.5 Required Point-in-Time Features Unavailable	18
39 Troubleshooting: Update Model Failed	19
40 Troubleshooting: Start Live Failed	19
41 Research Details: Point-in-Time Discipline	19
42 Research Details: Overlapping Horizons	19
43 Research Details: Block Bootstrap	19
44 Research Details: Adaptive Residual	20

45 Research Details: Drift	20
46 Research Details: Model Freshness and Retraining	20
47 Research Details: Retraining Contracts	20
48 Research Details: Model Selection	21
49 Using Research Mode	21
50 A Disciplined Model Experiment	21
51 What FinCompass Does Not Claim	22
52 Practical Checklist Before Relying on a Forecast	22
53 Worked Example: A New Supported Ticker	22
53.1 Step 1: Identify the Security	22
53.2 Step 2: Check Local History	23
53.3 Step 3: Select the Model Family	23
53.4 Step 4: Produce the Forecast	23
53.5 Step 5: Start Live	23
54 Worked Example: Interpreting a 61 Percent Forecast	23
55 Worked Example: Limited Evidence	23
56 Worked Example: Model Update Recommended	24
57 Worked Example: DCF Sensitivity	24
58 Worked Example: Risk Metrics Together	24
59 Worked Example: Factor Exposure	25
60 Research Workflow: Comparing Model Families	25
61 Research Workflow: Adding a New Feature	26
62 Research Workflow: Evaluating Calibration	26
63 Research Workflow: Reviewing a Retrained Candidate	26
64 Frequently Asked Questions	27
64.1 Why does FinCompass sometimes refuse to forecast a ticker?	27
64.2 Why can a model be Limited evidence but still available?	27
64.3 Why not train a neural network for every ticker?	27

64.4 Why use a pooled model instead of one model per stock?	27
64.5 Why does Live sometimes show no change?	27
64.6 Why can model freshness be old while ticker data are current?	27
64.7 Why is a model update not automatic?	27
64.8 Can I use the DCF output as the Forecast target?	27
64.9 Can I average probabilities across my portfolio?	28
65 Operational Checklist After Installation	28
66 Audit Checklist for a Release Build	28
67 Supported Horizons in Plain Language	28
68 Local-First Operating Assumptions	29
69 Disclaimer	29
A Glossary	29
B Reference Formulas	30
C Recommended Interpretation Language	31

1 What FinCompass Is

FINCOMPASS is a local-first financial research workbench. It combines four kinds of analysis that are intentionally kept separate:

1. an evidence score that summarizes currently observed company and market information;
2. a probabilistic Forecast for a precisely defined future event;
3. a Live workspace that tracks the frozen Forecast and may apply a separately governed adaptive adjustment when enough evidence has matured;
4. deterministic quantitative analytics such as performance, risk, ratios, DCF, bond mathematics, options, portfolio measures, factors and technical indicators.

Keep the four outputs apart in your head. A high evidence score is not a high chance of beating the index next year. A DCF value is a scenario, not a probability. A Sharpe ratio describes a history; it does not forecast one.

2 The Normal Workflow

For a supported security, the intended user journey is short:

Enter ticker → FinCompass identifies the market → obtains or checks local data → selects the strongest applicable pooled model → Forecast → Start Live.

You do not train a private model for every new ticker. Pooled model families are designed to cover defined market domains. For example, a US common-stock family model may be applicable to a newly entered US common stock even if that ticker was not one of the securities used to fit the model.

If local data are missing, FINCOMPASS should ask for an update and continue. If the family model is old, the software may recommend model maintenance, but an existing hard-valid applicable Forecast should remain available unless a documented hard-expiry rule is triggered.

3 First Use: Entering a Ticker

Enter a ticker in the Forecast workspace. The software performs several steps internally:

1. resolves the symbol and security identity;
2. classifies the asset class and market region;
3. chooses a benchmark policy;
4. checks whether local price history is sufficient;
5. searches the model registry for the strongest applicable model at the requested horizon;
6. checks scientific applicability, not merely whether the code can calculate features;
7. produces the Forecast when the required conditions are satisfied.

The Guided screen is supposed to speak in readiness language, not engineering internals. Model IDs, dataset hashes, locked-test thresholds and feature contracts belong in Research Details.

4 Choosing a Forecast Period

The forecast period determines the event being estimated. A six-month forecast and a twelve-month forecast are not interchangeable.

Suppose the displayed event is:

Probability that the security outperforms the S&P 500 over the next 12 months.

A value of 58% means the model estimates a 0.58 probability of that exact event under its declared data and applicability contract. It does not mean:

- the security will rise 58%;
- the model is 58% accurate;
- the security has a 58% chance of making money on every path;
- the expected return is 58%;
- the result is a recommendation.

5 Understanding the Benchmark

Many FINCOMPASS Forecasts are relative to a benchmark. If the benchmark is the S&P 500, the event compares the security's future return with the S&P 500 over the same horizon.

A security can decline in absolute terms and still outperform its benchmark. Conversely, it can rise and still underperform a stronger market. Always read the benchmark line before interpreting the probability.

6 Reading the Forecast Card

A Forecast card should answer five questions immediately:

1. **What is the probability?**
2. **What event is being forecast?**
3. **What is the horizon?**
4. **What benchmark is used?**
5. **How strong is the supporting evidence?**

A typical card might read:

54%

Estimated probability of outperforming the S&P 500 over the next year.

Evidence: Limited

The evidence label matters. It tells you whether the probability comes from a hard-valid Bayesian baseline or from a model that has passed stronger out-of-sample validation.

7 Evidence Labels

7.1 Limited Evidence

A Limited-evidence Forecast comes from a Bayesian baseline that passed hard validity checks but did not establish the stronger predictive skill required for the validated-research tier.

This means the model is usable as a probability reference but should be interpreted cautiously. It is not being presented as demonstrated alpha.

7.2 Research Validated

A research-validated model passed the configured locked-test statistical and stability gates on its declared historical research corpus. This is stronger evidence than a baseline, but the result remains limited to the data, market, horizon and applicability domain recorded in the model manifest.

7.3 Market Validated

This is the strongest model tier. In addition to the statistical gate, it requires stronger historical-universe and point-in-time controls such as survivorship handling, delistings and corporate-action treatment.

8 Why a Weak Model Can Still Produce a Probability

A model can be mathematically valid without showing meaningful incremental predictive skill. For example, it may converge, remain leakage-free, use a proper chronological split and produce reasonably calibrated probabilities, yet perform similarly to the historical event rate.

The correct conclusion is not necessarily “the software failed.” It is that the probability has limited demonstrated predictive evidence. FINCOMPASS separates these two judgments so the stronger validation thresholds do not need to be weakened simply to make the product usable.

9 Start Live

After a Forecast is available, choose **Start Live**. The same button can lead to two scientifically different modes, selected automatically by the software.

9.1 Tracking-Only Live

When the Forecast comes from a Limited-evidence Bayesian baseline, Live tracks the frozen probability and freshness state. It does not activate adaptive learning and does not apply a live residual. The current forecast therefore remains equal to the original forecast unless a different model is explicitly selected later.

This mode is still useful. It gives one place to monitor the Forecast, data recency and model status without overstating the evidence.

9.2 Adaptive-Eligible Live

A research-validated or market-validated model may be eligible to serve as the anchor for the adaptive Live layer. Even then, the adaptive adjustment is not automatically applied. The Live gate must accumulate

enough matured outcomes, enough distinct dates, enough elapsed time, acceptable probability losses, acceptable calibration and no active drift warning.

Until those conditions are met, the frozen Forecast remains in control.

10 How to Read Live

The most important Live values are:

Original forecast

The frozen anchor probability when Live was started.

Current forecast

The probability currently being applied. It may equal the original forecast when the adaptive layer is inactive.

Candidate forecast

A provisional adaptive estimate calculated from fresh context. It is not necessarily applied.

Evidence status

Whether the adaptive layer is warming, active or degraded.

A healthy display might say:

Current forecast: 61%

Original forecast: 62%

New information suggests a small decrease. Live adjustment is not being applied yet because FinCompass is still collecting enough completed outcomes.

That wording is more meaningful than exposing raw gate names or model-state codes.

11 Data Updates

FINCOMPASS keeps a durable local research corpus. Updating does not require downloading the entire history each time.

A normal update:

1. starts shortly before the most recent retained date;
2. downloads a short overlap and the missing tail;
3. compares overlapping values;
4. records provider revisions;
5. appends genuinely new observations;
6. preserves raw source frames and hashes where configured.

If the provider changes the price basis, such as raw versus adjusted history, FINCOMPASS should rebuild the affected symbol instead of appending incompatible prices.

12 Three Kinds of Freshness

The term “freshness” is easy to misuse. FINCOMPASS should distinguish three separate concepts.

12.1 Instrument-Data Freshness

Are the ticker and benchmark histories current enough to build today’s features?

12.2 Model-Training Freshness

How old was the family training corpus when the model was fitted?

12.3 Retrainability

Have enough new family observations and matured labels accumulated to justify or support a retrain?

A new ticker having data through today does not itself make a pooled family model stale. The model was trained on a family corpus, not on that newly entered ticker alone.

13 Model Maintenance

Model maintenance is separate from obtaining a Forecast. If an applicable hard-valid model already exists, an update may be recommended while the current Forecast remains available.

When an update is launched, FINCOMPASS should:

1. keep the current model intact;
2. rebuild the training dataset from the accumulated local corpus;
3. include only matured forward outcomes;
4. train a new immutable candidate;
5. evaluate the candidate;
6. compare the newer model with the current model;
7. require an explicit replacement action for a stronger active model.

If the candidate is not ready, fails computationally, or does not earn the intended evidence tier, the current Forecast should remain available.

14 Why Labels Must Mature

Assume a Forecast horizon of one year. An observation made today cannot have a one-year outcome tomorrow. The software may use today’s information to create a new Forecast, but that observation cannot become a supervised training label until the one-year endpoint actually occurs.

This rule prevents a major source of leakage. It also explains why adaptive Live learning can take time: each pending observation must wait for its original horizon to mature.

15 The Evidence Workspace

The evidence engine summarizes current information across five pillars:

- Quality;
- Financial Durability;
- Safety;
- Valuation;
- Cycle.

Each pillar can include uncertainty and evidence coverage. Missing data should move the result toward neutral and widen uncertainty rather than being silently replaced by optimistic assumptions.

The evidence score is not a forward return probability. It is a structured research summary.

16 Performance Analytics

FINCOMPASS includes deterministic historical-performance measures.

16.1 Annualized Return

For periodic returns r_t and m periods per year,

$$R_{ann} = \left(\prod_{t=1}^T (1 + r_t) \right)^{m/T} - 1. \quad (1)$$

Annualization makes histories of different lengths easier to compare, but it does not imply the same return will continue.

16.2 Volatility

Annualized volatility is approximately

$$\sigma_{ann} = s_r \sqrt{m}. \quad (2)$$

It summarizes dispersion of historical returns, not direction.

16.3 Sharpe Ratio

The Sharpe ratio compares average excess return with total volatility. A higher historical Sharpe is not a guarantee of a better future outcome.

16.4 Sortino Ratio

Sortino focuses on downside deviation rather than all volatility. It can be useful when upside and downside variability should not be treated symmetrically.

16.5 Maximum Drawdown

Maximum drawdown is the largest peak-to-trough decline in the historical wealth path. It is a historical observation, not a worst-case bound.

16.6 Beta and Tracking Error

Beta describes historical sensitivity to a benchmark. Tracking error describes dispersion of active returns relative to the benchmark. Neither is itself a forecast.

17 Risk Analytics

17.1 Value at Risk

VaR estimates a loss threshold under a stated model and confidence level. A 95% one-day VaR is not the maximum possible one-day loss. Losses can exceed VaR.

17.2 Expected Shortfall

Expected shortfall summarizes average loss beyond a VaR threshold under the historical or assumed distribution. It provides tail information that VaR alone does not.

17.3 EWMA Volatility

Exponentially weighted volatility gives more weight to recent observations, allowing measured volatility to respond faster to changing market conditions.

18 Technical Analytics

Technical calculations include moving averages, exponential moving averages, RSI, MACD, Bollinger Bands and ATR.

These can help describe trend, momentum and volatility. They do not automatically become Forecast features. An indicator must be separately admitted through the feature-governance process before it can influence the forecasting model.

19 Financial Statements and Normalization

Data providers use different field names and conventions. FINCOMPASS converts provider-specific vocabulary into canonical income-statement, balance-sheet and cash-flow fields.

Every normalized value should carry enough context to identify whether it was reported directly, mapped from a provider alias, derived from other fields, unavailable or not applicable. Missing values remain missing. They are not silently converted to zero.

This matters because a missing inventory balance is not the same thing as zero inventory, and missing debt is not the same thing as no debt.

20 Financial Ratios

Ratios are grouped into several families.

20.1 Profitability

Examples include gross margin, operating margin, net margin, return on assets and return on equity. Ratios with negative or structurally invalid denominators require careful interpretation.

20.2 Liquidity

Current ratio and quick ratio compare near-term resources with near-term obligations. Sector context matters: a bank and a manufacturer should not be interpreted using identical balance-sheet intuition.

20.3 Leverage

Debt-to-equity, debt-to-assets and interest-coverage measures describe capital structure and debt burden. Negative equity can make some common leverage ratios economically difficult to interpret; the software should not hide that problem by returning a convenient finite value.

20.4 Efficiency

Asset turnover, inventory turnover and receivables measures describe how efficiently resources are used. Period-average balance-sheet denominators are often preferable to a single ending balance when appropriate data are available.

21 Discounted Cash Flow Valuation

DCF estimates present value under explicit assumptions. The unlevered free-cash-flow identity is

$$FCFF = EBIT(1 - T) + D\&A - CapEx - \Delta NWC. \quad (3)$$

Projected FCFF values are discounted at WACC. Terminal value may use perpetual growth or an exit multiple.

The output is only as reliable as the assumptions. A useful DCF workflow therefore emphasizes sensitivity analysis:

- revenue growth;
- operating margin;
- tax rate;
- reinvestment;
- WACC;
- terminal growth or exit multiple;
- net debt and diluted shares.

Do not treat one DCF number as truth. Compare scenarios and inspect which assumptions dominate the result.

22 Fixed-Income Analytics

The fixed-income module calculates bond price, current yield, yield to maturity, Macaulay duration, modified duration, convexity and DV01 from explicit terms.

22.1 Bond Price

For face value F , coupon C , yield y , frequency m and n remaining periods,

$$P = \sum_{k=1}^n \frac{C/m}{(1 + y/m)^k} + \frac{F}{(1 + y/m)^n}. \quad (4)$$

22.2 Duration

Macaulay duration is a weighted average time to cash flow. Modified duration approximates percentage price sensitivity to a small yield change. Convexity improves the approximation for larger changes.

These are interest-rate sensitivity measures, not guaranteed future price changes.

23 Options Analytics

European Black-Scholes-Merton calculations include theoretical call/put value and major Greeks.

Delta

First-order sensitivity to the underlying price.

Gamma

Change in Delta as the underlying changes.

Vega

Sensitivity to volatility.

Theta

Sensitivity to passage of time.

Rho

Sensitivity to interest rates.

Actual option prices can differ because of volatility surfaces, liquidity, discrete dividends, early exercise and market microstructure. The calculation is a model result under stated assumptions.

24 Portfolio Analytics

Portfolio analytics combine explicit holding weights with historical returns or a covariance matrix.

For weight vector w and covariance matrix Σ ,

$$\sigma_p^2 = w^\top \Sigma w. \quad (5)$$

Euler risk contributions can decompose portfolio volatility into holding-level contributions.

A portfolio probability should not be created by simply averaging the Forecast probabilities of its holdings. The joint distribution and dependence structure matter.

25 Factor and Regression Analytics

OLS factor models estimate descriptive exposures. A single-factor model may be written

$$r_i - r_f = \alpha + \beta(r_m - r_f) + \epsilon. \quad (6)$$

A multi-factor model extends the right-hand side with additional factor returns.

The output includes coefficients, standard errors, t-statistics and fit measures. A statistically significant loading does not mean a factor tilt will generate future returns. In-sample exposure is not a forecast.

26 The Bayesian Reference Model

The standard Forecast baseline uses regularized Bayesian logistic regression. For feature vector x ,

$$P(Y = 1 | x) = \frac{1}{1 + \exp[-(\beta_0 + x^\top \beta)]}. \quad (7)$$

Shrinkage priors pull coefficients toward zero when the data do not strongly support large effects. This reduces instability and makes the model a useful reference against which more complex models can be judged.

The implementation approximates posterior coefficient uncertainty around the fitted mode and propagates that uncertainty into a probability range.

27 Calibration in Practical Terms

If a system repeatedly assigns 60% probability to comparable events, a well-calibrated system should see those events occur approximately 60% of the time over a sufficiently large and stable sample.

Calibration is therefore different from ranking. A model can be good at ranking companies from stronger to weaker while giving overconfident probabilities. Another model can have modest ranking power but reasonably calibrated probabilities.

FINCOMPASS evaluates probability quality using proper scores and calibration diagnostics rather than relying on AUC alone.

28 Brier Score and Log Loss

For one binary outcome y and probability p ,

$$BS = (p - y)^2. \quad (8)$$

Lower is better. Brier skill compares the model with a reference forecast such as the historical event rate.

Log loss is

$$LL = -[y \log p + (1 - y) \log(1 - p)]. \quad (9)$$

It penalizes highly confident wrong predictions more strongly.

These metrics are appropriate because the model is trying to produce probabilities, not merely class labels.

29 ROC AUC

ROC AUC measures ranking. AUC of 0.5 corresponds to random ranking in a balanced conceptual sense; values above 0.5 indicate better ordering of positive and negative outcomes.

AUC is useful but insufficient. A high-AUC model can be badly calibrated, and a probability product must care about both ranking and probability accuracy.

30 Why the Test Is Locked

The locked test is meant to represent data not used in model design. If you repeatedly change features after seeing the locked-test result, the test is no longer locked. It has become another development set.

FINCOMPASS therefore separates:

- training;
- validation/calibration;
- ensemble-weight selection where applicable;
- final calibration;
- locked testing.

Chronological purging and embargo rules further reduce leakage from overlapping forward horizons.

31 Regime-Aware Bayesian Research Model

Markets can behave differently during stress, neutral and risk-on environments. The regime-aware model introduces a hidden market state estimated from benchmark-only, backward-looking features.

A three-state Gaussian hidden Markov model estimates filtered probabilities for the states. These probabilities are appended to the Bayesian reference features. Later test observations are processed forward so future data cannot rewrite earlier state assignments.

The regime model is not assumed to be better. It must demonstrate incremental value. On the current reference corpus, some horizons improved selected metrics while others worsened. The model therefore remains a research alternative rather than the default.

32 Enhanced Model

The enhanced model combines Bayesian logistic regression with nonlinear learners such as histogram gradient boosting and random forest. Separate chronological stages are used for component calibration, ensemble weighting and final calibration.

The important principle is not the exact list of algorithms. The principle is that complexity must improve on the simpler reference under protected out-of-sample evaluation. If it does not, the simpler model should remain preferred.

33 Current Reference Evidence

The current bundled research evidence uses a small historical US-equity sample and the S&P 500. It is useful for demonstrating software behavior and the validation framework but is not a survivorship-

complete market dataset.

Table 1: Bayesian reference results on the declared locked tests.

Horizon	Test n	AUC	Brier skill	ECE
6 months	539	0.419	-0.023	0.056
12 months	532	0.480	-0.012	0.035
24 months	518	0.415	-0.004	0.064
36 months	497	0.303	-0.108	0.199

These models are retained as Limited-evidence baselines. They are not promoted to stronger tiers merely because the software can fit them.

The stronger twelve-month enhanced model recorded approximately AUC 0.576, Brier skill 0.036, log-loss skill 0.027 and ECE 0.045 on its declared locked test. That supports its research-validation tier within the limits of the corpus.

34 Applicability Checks

Before trusting a Forecast, verify that the model’s declared domain matches the security:

- asset class;
- region;
- security type;
- benchmark family;
- forecast horizon;
- feature frequency;
- feature contract.

A US common-stock model should not silently forecast a bond, commodity or cryptoasset simply because the software can calculate returns for both.

35 Model Registry and Integrity

Each model has an immutable artifact and manifest. The manifest records the model hash, target, features, settings, dataset provenance, split metadata, evidence tier and applicability domain.

A current model is selected explicitly. A newer candidate does not replace it merely because it has a later timestamp. This prevents accidental “newest model wins” behavior.

36 Candidate Replacement

When a model is retrained, the new model records its parent. The current model remains intact while the candidate is evaluated.

For Guided use, the comparison should focus on meaningful fields such as:

- market data through date;
- forecast period;
- benchmark;
- evidence tier;
- freshness.

Detailed Brier, AUC and calibration metrics belong in Research Details.

37 What Happens When an Update Fails

An unsuccessful optional model update should not remove a current hard-valid Forecast. The expected behavior is:

The newer model was not ready, so FinCompass kept the current model. Your forecast remains available.

Technical diagnostics remain available for research inspection, but the normal workflow should recover automatically.

38 Troubleshooting: Forecast Unavailable

There are several distinct reasons a Forecast can be unavailable.

38.1 Not Enough Ticker History

The current security does not have enough retained data to build the model's required features. Update local data and try again.

38.2 Benchmark History Missing

The security may be available while its benchmark is missing. Update both histories.

38.3 No Applicable Model Family

The security is outside the declared domain of installed model families. Analytics may still be available. This is a scientific limitation, not a software crash.

38.4 Model Artifact Failed Integrity Checks

A model file may be missing or its hash may not match the manifest. Reinstall or restore the correct artifact rather than bypassing the integrity check.

38.5 Required Point-in-Time Features Unavailable

Some models may require additional point-in-time fundamental features. If those cannot be reconstructed under the declared data contract, the model should not run.

39 Troubleshooting: Update Model Failed

Model maintenance can fail for different reasons:

- the declared retraining recipe is unsupported;
- the local corpus lacks enough target or benchmark history;
- labels have not matured;
- the trainer encounters a computational failure;
- the candidate is hard-invalid;
- the candidate is hard-valid but does not earn the stronger evidence tier.

These cases should be distinguished in Research Details. In Guided use, the current model should remain available whenever it was already hard-valid.

40 Troubleshooting: Start Live Failed

If the selected model is a Bayesian baseline, Live should start in tracking-only mode without trying to activate adaptive control. If the selected model is research-validated or market-validated, it may require explicit activation before adaptive Live is eligible.

If Live cannot start, check whether the problem is model applicability, artifact integrity, benchmark data, or a missing current Forecast. Do not resolve the issue by promoting an ineligible baseline to adaptive control.

41 Research Details: Point-in-Time Discipline

Point-in-time discipline means that every feature value must have been available at the observation date. This is straightforward for historical prices and more subtle for fundamentals and macro data.

A financial statement belongs to the feature history on its filing/publication date, not automatically on the fiscal period end. Revised macro data require similar care if a historical backtest claims real-time availability.

42 Research Details: Overlapping Horizons

Long-horizon labels overlap. Monthly observations with a one-year target share much of the same future return path. Treating them as independent rows understates uncertainty.

This is why temporal breadth, unique observation dates, walk-forward stability and block-bootstrap methods matter. A dataset with many rows can still have limited independent temporal information.

43 Research Details: Block Bootstrap

The moving-date-block bootstrap resamples contiguous date blocks and keeps same-date cross-sectional observations together. This preserves part of the serial and cross-sectional dependence that an iid row bootstrap would destroy.

Bootstrap intervals do not prove future robustness, but they provide a more realistic measure of uncertainty around the observed test statistics.

44 Research Details: Adaptive Residual

The Live residual adjusts the anchor in log-odds space:

$$\text{logit}(p_{\text{candidate}}) = \text{logit}(p_0) + \delta(z_t). \quad (10)$$

The correction is bounded. When the gate is inactive, the applied probability returns to p_0 even if a candidate probability can be calculated.

This design prevents the fresh-event layer from silently redefining the validated anchor model.

45 Research Details: Drift

Distribution shift can make a previously useful adaptive relationship unreliable. The Live layer therefore monitors anchor-relative performance and calibration over matured outcomes. A drift alert disables the applied residual and returns control to the frozen anchor.

Drift monitoring does not guarantee detection of every regime change. It is a defensive mechanism, not proof that the future distribution is stable.

46 Research Details: Model Freshness and Retraining

A model manifest should store the training cutoff and the data/feature contract. Retraining should ask whether enough new family observations and matured labels exist, not merely whether the currently viewed ticker has newer prices.

This distinction is particularly important for pooled models. The user's ticker supplies today's feature vector; it does not automatically define the family retraining corpus.

47 Research Details: Retraining Contracts

A model should declare exactly how it can be retrained. A robust contract contains:

- trainer family;
- recipe identifier;
- feature contract;
- benchmark family;
- horizon;
- retrain support flag.

The software should never infer a training recipe from a human-readable profile name.

48 Research Details: Model Selection

Model selection should prefer the strongest applicable tier, not the newest file. Within the same tier and domain, additional tie-breaking rules can consider creation date, exact feature compatibility and artifact integrity.

A model from the wrong region or benchmark family is not a fallback. It is inapplicable.

49 Using Research Mode

Research mode exposes the mechanics hidden from Guided use. Typical tasks include:

- inspect model manifests and evidence tiers;
- review locked-test metrics;
- compare Bayesian reference and enhanced models;
- inspect calibration and uncertainty;
- examine experiment history;
- review candidate lineage;
- inspect data coverage and provenance;
- run controlled retraining experiments;
- inspect Live source health and adaptive state.

Research mode is not a bypass around validation. It exposes more information; it does not weaken the scientific controls.

50 A Disciplined Model Experiment

When testing a new feature family or algorithm:

1. define the target and benchmark before examining results;
2. define point-in-time feature semantics;
3. define missing-data handling;
4. use chronological development splits;
5. tune only inside development data;
6. freeze the candidate;
7. evaluate the locked test once;
8. compare with the unconditional and Bayesian reference models;
9. retain the simpler model if the extra complexity does not demonstrate robust improvement;
10. record the experiment even when it fails.

This process reduces the risk of discovering “performance” that exists only because many alternatives were tried.

51 What FinCompass Does Not Claim

FINCOMPASS does not claim:

- guaranteed returns;
- guaranteed outperformance;
- a universal model for every ticker and market;
- that a Limited-evidence probability has demonstrated alpha;
- that a research-validated result is market-grade proof;
- that a DCF value is a target price;
- that VaR is the worst possible loss;
- that historical factor exposure is causal;
- that a backtest will repeat in the future.

52 Practical Checklist Before Relying on a Forecast

Ask:

1. What exact event is being forecast?
2. What is the horizon?
3. What benchmark is used?
4. Does the security match the model's applicability domain?
5. Is the ticker data current enough to construct features?
6. How old is the model's training corpus?
7. Is the evidence Limited, research validated or market validated?
8. Is Live tracking-only or adaptive eligible?
9. If an update failed, did the current model remain intact?
10. Are you interpreting the result as research evidence rather than a guarantee?

53 Worked Example: A New Supported Ticker

Assume you enter a valid US common-stock ticker that is not one of the securities used to fit the bundled reference model.

53.1 Step 1: Identify the Security

FinCompass resolves the symbol, security type, exchange/region and benchmark family. If the security is classified as a US common stock, the resolver can consider the US large-cap/common-equity family when the rest of the domain matches.

53.2 Step 2: Check Local History

The software checks whether enough ticker and benchmark history exists to calculate the current feature vector. If not, choose **Update data**. The update should fetch the missing tail plus an overlap window, then automatically return to the Forecast flow.

53.3 Step 3: Select the Model Family

FinCompass searches the installed model registry at the selected horizon. It prefers the strongest applicable tier. If a research-validated model exists for the same domain and horizon, it is selected before a Limited-evidence Bayesian baseline.

53.4 Step 4: Produce the Forecast

The current ticker's feature vector is calculated under the same feature contract used to train the pooled family. The ticker does not need to be retrained individually merely because it is new to the user.

53.5 Step 5: Start Live

If the selected model is a baseline, Live starts in tracking-only mode. If it is a stronger validated model, the software can use the adaptive-eligible path, subject to explicit activation and the Live gate.

54 Worked Example: Interpreting a 61 Percent Forecast

Suppose FinCompass shows:

61% probability of outperforming the S&P 500 over the next 12 months. Evidence: Research validated.

A careful interpretation is:

- the model estimates the defined relative-return event at 0.61;
- the estimate is conditional on the current feature vector and model contract;
- the evidence tier indicates the model passed the configured research gate on its declared locked test;
- the probability is not a guarantee and is not a target return;
- the model can still be wrong on this specific security and date.

Do not convert 61% into a position size automatically. Position sizing requires a separate decision framework that considers expected payoff, loss severity, portfolio concentration, liquidity and personal constraints.

55 Worked Example: Limited Evidence

Suppose the 24-month horizon shows:

52% probability of outperforming the S&P 500 over the next 24 months. Evidence: Limited.

The right interpretation is that the Bayesian reference model is hard-valid for the domain but has not demonstrated enough out-of-sample improvement to support a stronger claim. The number can serve as a reference probability, but it should carry less decision weight than a stronger validated model.

If Live is started, it should track the frozen 52% rather than applying adaptive learning.

56 Worked Example: Model Update Recommended

Assume the current model was trained several years ago and FinCompass reports that newer family data are available.

You should still be able to view the current Forecast if the model remains hard-valid. The maintenance path is separate:

1. choose Update model;
2. the current model remains unchanged;
3. FinCompass rebuilds the accumulated family dataset;
4. only matured outcomes become labels;
5. a new candidate is trained and evaluated;
6. if the candidate is stronger and eligible, you can compare it with the current model;
7. replacement occurs only through the governed selection action.

If the update does not complete, the current Forecast should remain available.

57 Worked Example: DCF Sensitivity

Suppose a DCF produces a base-case per-share value of 120 under WACC 9% and terminal growth 3%. That value is not meaningful without sensitivity.

A simple grid might show:

	$g = 2\%$	$g = 3\%$	$g = 4\%$
WACC 8%	132	145	162
WACC 9%	111	120	132
WACC 10%	95	102	111

The key insight is not the single 120 value. It is how strongly the valuation changes when plausible assumptions change. If a small change in WACC or terminal growth produces a large swing, the valuation is assumption-sensitive and should be treated accordingly.

58 Worked Example: Risk Metrics Together

Assume a security has:

- annualized volatility 35%;
- maximum drawdown -48%;

- beta 1.4;
- Sharpe ratio 0.55;
- historical 95% one-day VaR of 3.2%.

These numbers tell a coherent historical story: the security has been volatile, has experienced a large peak-to-trough loss, and has moved more than the market on average. The Sharpe ratio indicates only moderate historical return per unit of volatility. VaR summarizes a historical loss threshold but does not cap extreme losses.

None of these metrics says whether the security will outperform over the next year. That is why they are kept separate from the Forecast probability.

59 Worked Example: Factor Exposure

Suppose a multi-factor regression reports a market beta of 1.25, a positive momentum loading and a statistically significant intercept over the sample.

A disciplined interpretation is:

- the security historically moved more than the market in the sample;
- its returns co-varied with the supplied momentum factor;
- the intercept is an in-sample estimate conditioned on the factor specification;
- statistical significance does not prove future alpha;
- factor returns supplied to the regression must themselves be trustworthy and aligned in time.

60 Research Workflow: Comparing Model Families

When comparing Bayesian reference, regime-aware Bayesian and enhanced models, build a table containing at least:

- test sample size and unique dates;
- Brier score and skill;
- log loss and skill;
- ROC AUC and average precision;
- ECE and calibration slope;
- bootstrap uncertainty;
- walk-forward stability;
- model complexity;
- applicability domain;
- data-quality limitations.

Prefer the simpler model when the more complex model does not demonstrate robust incremental improvement. Complexity is a cost that should be justified by evidence.

61 Research Workflow: Adding a New Feature

Before adding a new ratio, technical indicator, factor or macro variable to Forecast:

1. write a precise formula and units;
2. define the observation timestamp and point-in-time availability;
3. define missing-value handling;
4. determine whether revisions/restatements can change historical values;
5. register the feature contract;
6. test it inside chronological development data;
7. compare with the existing Bayesian reference;
8. protect the locked test from repeated search;
9. retain the feature only if the evidence justifies it.

62 Research Workflow: Evaluating Calibration

Calibration should be inspected numerically and visually. Useful questions include:

- do 50–60% predictions occur at roughly that frequency?;
- are high-confidence forecasts systematically too extreme?;
- does calibration change across time periods?;
- is one market regime responsible for most calibration error?;
- is the calibration sample large enough to support the chosen calibrator?;
- does recalibration improve proper scores without hiding poor ranking?;

A calibrated probability can still have weak skill. Calibration and discrimination should be reported together.

63 Research Workflow: Reviewing a Retrained Candidate

A candidate comparison should answer:

1. Was the dataset extended correctly?
2. How many new matured labels were added?
3. Did the feature contract change?
4. Did the benchmark or applicability domain change?
5. Did proper-score skill improve?
6. Did calibration improve or deteriorate?
7. Is the improvement stable across walk-forward windows?

8. Is the candidate the same evidence tier, an upgrade, or a downgrade?
9. Does the candidate introduce new data-quality limitations?

A newer timestamp is not a sufficient reason to replace the current model.

64 Frequently Asked Questions

64.1 Why does FinCompass sometimes refuse to forecast a ticker?

Because an appropriate model family may not exist for that market, security type, benchmark family or horizon. Returning no probability is better than applying a model outside its scientific domain.

64.2 Why can a model be Limited evidence but still available?

Because hard validity and predictive skill are different. The model can be coherent and calibrated enough to serve as a reference while failing the stronger out-of-sample skill threshold.

64.3 Why not train a neural network for every ticker?

More complex models require more data, can be harder to calibrate, and do not automatically generalize better. FinCompass uses a simple Bayesian reference so every complex candidate has a clear benchmark to beat.

64.4 Why use a pooled model instead of one model per stock?

A single stock often provides too little independent long-horizon data. Pooling across a defined family can improve estimation while retaining a clear applicability contract.

64.5 Why does Live sometimes show no change?

Because the adaptive gate may be warming, the source data may be stale, or the selected model may be baseline-only. In those cases the frozen anchor remains the scientifically appropriate value.

64.6 Why can model freshness be old while ticker data are current?

Because the model was trained on a family corpus at an earlier cutoff, while today's ticker features are calculated from current ticker data. These are different concepts.

64.7 Why is a model update not automatic?

Because retraining can produce a worse model. Preserving the current artifact and requiring governed replacement prevents silent degradation.

64.8 Can I use the DCF output as the Forecast target?

No. DCF is a valuation scenario based on cash-flow and discount-rate assumptions. Forecast is an empirical probability for a defined future event. They can be viewed together but should not be conflated.

64.9 Can I average probabilities across my portfolio?

Not as a valid portfolio forecast. The outcomes are dependent and the portfolio target must be defined separately.

65 Operational Checklist After Installation

After installing or moving the application to a new machine:

1. confirm the application starts without a console dependency;
2. check that the bundled model registry is visible;
3. enter a known supported ticker and produce a Forecast;
4. enter a different supported ticker that was not in the training symbols;
5. update local data if prompted;
6. start Live;
7. close and reopen the application;
8. verify that local data, settings and model state remain coherent;
9. only then begin Model Lab experiments.

66 Audit Checklist for a Release Build

A release should not be called complete solely because unit tests pass. Confirm:

- actual packaged new-ticker Forecast works;
- baseline Start Live works in tracking-only mode;
- validated-model Start Live follows the adaptive path;
- an optional model update failure does not destroy the current Forecast;
- every advertised horizon has an honest retrain-support status;
- no stale project-management terminology appears in user-facing text;
- the technical manuscript and guide match the implemented behavior;
- one canonical source exists for each document;
- no temporary LaTeX build logs or staging copies are distributed as documentation.

67 Supported Horizons in Plain Language

The Guided product currently talks about four forecast periods. Each period is a different event. Do not treat a six-month probability as a twelve-month probability with a different label.

Horizon	How to think about it
6 months	Nearer event; more independent labels over a given calendar span, but more noise.
12 months	The workhorse relative-outperformance event for the bundled US-equity family.
24 months	Longer event; fewer matured labels and more overlap. Interpret more cautiously.
36 months	Longest bundled event. Independent sample is thin. Treat Limited-evidence results as reference only.

A horizon may be forecast-ready without being retrain-ready. If the software offers **Update model** for a horizon, that horizon must have a working retraining contract. If it does not, the button should not appear.

68 Local-First Operating Assumptions

FINCOMPASS is designed to keep the research corpus and model artifacts on the machine that runs the application. That has three practical consequences.

1. Training never silently downloads a new universe behind your back. Data update is a separate, inspectable action.
2. Restart should restore the local corpus, the selected model and the current Forecast/Live state.
3. Moving the application to another computer is a data-and-artifact move, not a cloud-session resume.

69 Disclaimer

Nothing in this guide is investment advice, a solicitation, or a representation that any probability, valuation or risk number will be realised. Use the tool as a research workbench. Position sizing, tax, liquidity and suitability remain the user's responsibility.

A Glossary

Adaptive candidate

Provisional Live probability after applying the bounded event residual.

Anchor

Frozen Forecast probability from the selected forecast model.

Applicability domain

Declared conditions under which a model may be used.

Bayesian baseline

Hard-valid probability model that has not established stronger predictive skill.

Benchmark

Reference security or index used to define relative outperformance.

Brier score

Proper probability score equal to squared error between forecast probability and binary outcome.

Calibration

Agreement between stated probabilities and observed event frequencies.

Candidate model

Newly trained immutable model that has not replaced the current model.

Evidence tier

Strength-of-evidence classification attached to a forecast model.

Feature contract

Versioned definition of the model inputs and their semantics.

Locked test

Final chronological evaluation set not used for fitting or tuning.

Matured label

Forward outcome whose original forecast horizon has completed.

Model family

Pooled model designed for a declared market/domain rather than one ticker only.

Point-in-time

Rule that a feature must be available at the historical observation date.

Proper scoring rule

Loss function that rewards truthful probability forecasts in expectation.

Regime

Latent market-state representation such as risk-off, neutral or risk-on.

Retraining contract

Explicit manifest fields that identify how a model can be rebuilt.

Tracking-only Live

Live mode that monitors a baseline Forecast without adaptive learning.

B Reference Formulas

Measure	Core expression or interpretation
Bayesian forecast	$P(Y = 1 \mid x) = [1 + \exp(-\beta_0 - x^\top \beta)]^{-1}$
Brier score	$(p - y)^2$
Log loss	$-[y \log p + (1 - y) \log(1 - p)]$
Annualized volatility	$s_r \sqrt{m}$
Portfolio variance	$w^\top \Sigma w$
FCFF	$EBIT(1 - T) + D\&A - CapEx - \Delta NWC$
DCF enterprise value	$\sum FCFF_t / (1 + WACC)^t + TV / (1 + WACC)^N$
Bond price	Present value of coupons plus principal under stated yield convention

Measure	Core expression or interpretation
Live candidate	$\text{logit}(p_{\text{candidate}}) = \text{logit}(p_0) + \delta(z_t)$

C Recommended Interpretation Language

Situation	Preferred explanation
Limited evidence	“Forecast available. The Bayesian reference model is scientifically usable, but stronger out-of-sample improvement has not been established.”
Research validated	“The model passed the configured research validation gates for its declared data and applicability domain.”
Model update failed	“The newer model was not ready, so FinCompass kept the current model. Your forecast remains available.”
Baseline Live	“Live is tracking the frozen forecast. Adaptive changes are disabled for this evidence tier.”
Unsupported market	“FinCompass can analyze this instrument, but an appropriate forecasting family is not currently available.”